

Deliverable 2: Scalable Automated MLOps Pipeline

AETHER Atmospheric Intelligence Platform -- Developed by Nuralis Labs

1. Executive Overview

Deliverable 2 represents the continuous, automated MLOps pipeline architecture powering AETHER. The system employs automated workflows to handle high-frequency data ingestion, cloud feature store synchronization, automated model retraining, and model registry updates without human intervention.

2. Pipeline Architecture and Schedule

- **1. Hourly Feature Extraction Pipeline** (src/feature_pipeline.py): Executes every hour (cron: 0 * * * *) to ingest Open-Meteo REST telemetry, engineer 27 signals (lags, rolling averages, seasonal sines/cosines), and push updates to Hopsworks Cloud Feature Store (aqi_weather_fg v1).
- **2. Daily Model Retraining and Evaluation** (src/train.py): Executes every day at midnight (cron: 0 0 * * *). Trains multi-output Random Forest regressors, evaluates performance against Ridge and PyTorch MLP benchmarks, updates metrics.json, and registers winning artifacts to Hopsworks Model Registry.
- **3. 365-Day Historical Backfill Pipeline** (src/backfill.py): Manages 35,136 hourly observations across Karachi, Chicago, Sydney, and Vienna for cloud feature store initialization.

3. GitHub Actions CI/CD Automation

Pipeline automation is defined in .github/workflows/mlops_pipeline.yml using GitHub Actions serverless runners:

- **Workflow Setup:** Configured with workflow_dispatch and scheduled crons for hands-free automation.
- **Cloud Security:** Connects securely to Hopsworks Cloud via repository secrets (HOPSWORKS_API_KEY).
- **Execution Runner:** Automated build container executing Python 3.10, feature sync, and model evaluation.

4. Repository Artifacts

- **1. Workflow Spec:** .github/workflows/mlops_pipeline.yml
- **2. Feature Script:** src/feature_pipeline.py
- **3. Training Script:** src/train.py
- **4. Backfill Script:** src/backfill.py
- **5. Repository URL:** <https://github.com/M-Sohaib-Siddiqui/Aether-air-intelligence>

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